



Atmospheric Thermodynamics



Chapter 1: Introduction



Objective

- To Define atmospheric Thermodynamics
- To Know Thermodynamics system types
- To Identify Thermodynamic variables
- To Explain Fundamental Gas Laws

Introduction

- *Meteorology* is concerned with describing the present state of the atmosphere (temperature, pressure, winds, humidity, precipitation, cloud cover, etc.) and in predicting the evolution of these primary variables over time intervals of a few days.
- To do this it is important to *describe* the atmosphere in terms of its primary variables.

Cont. . .

❖ In general, meteorology may be categorized into three disciplines:

Physical meteorology

Dynamic meteorology

Synoptic meteorology

✓ In physical meteorology, it can be further categorized into

Cloud physics

Radiation

Thermodynamics

Cont. . .

- An essential tool needed in this description is **thermodynamics**, which helps relate the fundamental quantities of pressure, temperature and density as atmospheric parcels move from place to place.

Cont. . .

- Such parcels contract and expand, their temperatures rise and fall; water changes phase, back and forth from vapor to liquid to ice; chemical constituents react, etc.
- The key to understanding these changes lies in applications of the laws of thermodynamics which relate these changes to fluxes of energy.

Cont. . .

- The field of thermodynamics is concerned with the science of energy focusing on energy storage and energy conversion processes.
- As we may expose a mass to heating/cooling or to volumetric compression/expansion.
- During such processes we are transferring energy into or out of the mass, so it changes its conditions expressed by properties like temperature, pressure, and volume.

Cont. . .

- Thermodynamics plays an important role in our quantitative understanding of atmospheric phenomena ranging from the smallest cloud microphysical processes to the general circulation of the atmosphere.
- Thermodynamics deals with the transfer of energy from one place to another and from one form to another.

Atmospheric Thermodynamics Definition

Atmospheric Thermodynamics: Definition

- Atmospheric Thermodynamics is the branch of meteorology that studies the energy, heat, and temperature changes in the Earth's atmosphere.
- It explains how air parcels move, clouds form, weather patterns develop, and energy is transferred in the atmosphere.

Cont. . .

- It is the study of the physical and energetic processes governing the behavior of the atmosphere.
- It is a science that deals with heat and work and those properties of matter that relate to heat and work.
- It applies the laws of thermodynamics to understand how energy, heat, and moisture interact within the atmosphere, influencing weather and climate phenomena.

Cont. . .

- Atmospheric thermodynamics is the study of heat-to-work transformations (and their reverse) that take place in the Earth's atmosphere and manifest as weather or climate.
- Atmospheric thermodynamics use the laws of classical thermodynamics, to describe and explain such phenomena as the properties of moist air, the formation of clouds, atmospheric convection, boundary layer meteorology, and vertical instabilities in the atmosphere.

The Reason for learning Atmospheric Thermodynamics

- The atmosphere is made up of mixture of gases how this mixture behaves under spatially varying temperature and pressure is crucial to predicting the weather and climate.
- Therefore it is necessary for us to know the laws that govern the detail of these gases.

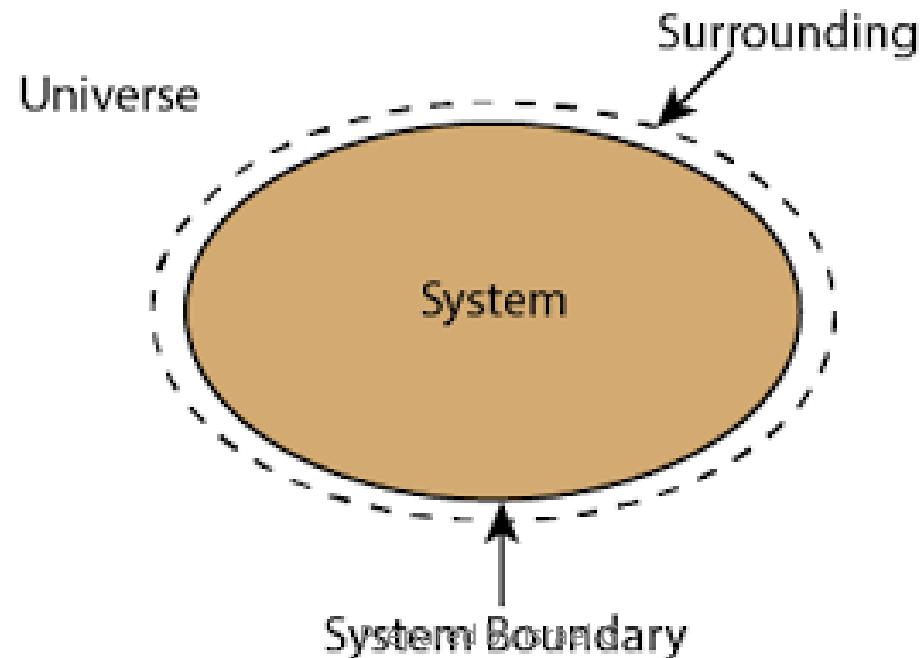
Thermodynamic Systems

Thermodynamic systems

- A thermodynamic system is a region of space or a set of materials that are under study, separated from its surroundings by a boundary.
 - Depending on how energy and matter interact across this boundary, thermodynamic systems are classified into three types:
 - while everything outside the system is called the **surroundings**.
- Systems are classified based on how they exchange energy and matter with their surroundings.

Thermodynamic systems

- ❖ **Surroundings or environment:** Everything external to the matter or space, which is under thermodynamic study is called surroundings or environment.
- ❖ **Boundary:** The boundary that separates the system and surrounding is called as system boundary. The system boundary may be fixed or moving.



Thermodynamic system types

Open System

- Both matter and energy (heat or work) to be exchanged with its surroundings.
- Example in Meteorology:
 - ✓ A hurricane is an open system because it continuously exchanges heat and moisture with the ocean and atmosphere. A cloud forming due to rising air also gains moisture from surrounding air masses.
 - ✓ The Earth's atmosphere is an open system because it exchanges heat with space (radiation) and receives matter (moisture, gases) from the surface.
 - ✓ A boiling pot of water—heat enters from the stove, and steam (matter) escapes into the air.

Closed System

Allows energy exchange (heat or work) but not matter exchange with its surroundings.

✓ Example in Meteorology:

✓ A parcel of air that expands and cools as it rises in the atmosphere is often treated as a closed system in thermodynamic calculations (assuming no mixing with surrounding air).

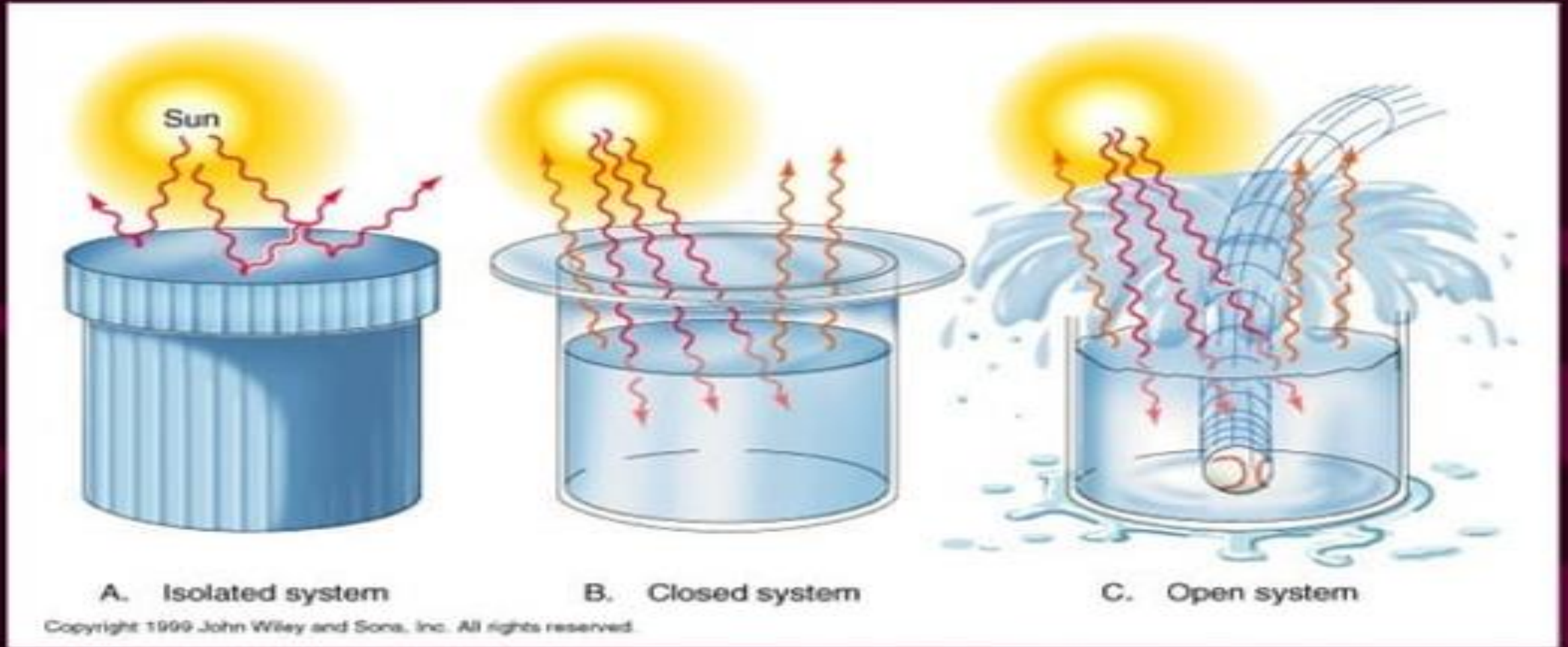
✓ A sealed weather balloon filled with gas—heat can transfer in and out, but no gas escapes or enters.

✓ A pressure cooker—heat enters and leaves, but steam is trapped inside unless the valve is opened.

Isolated System

- A system that does not exchange energy or matter with its surroundings.
- Example:
 - Example in Meteorology: The universe (in an ideal sense) is often considered an isolated system because it does not exchange energy or matter with anything outside it.
 - Example in Daily Life: A well-insulated thermos flask (in an ideal case)—it prevents heat from escaping or entering, and no liquid enters or leaves.

Cont. . .



Closed system: exchange of energy but negligible exchange of mass with surroundings

Thermodynamic Variables

Thermodynamic Variables or Parameters

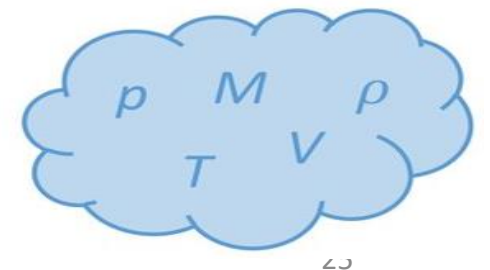
- ❖ Thermodynamic **variables** are the quantities like pressure, volume and temperature, which help us to study the behavior of a thermodynamic system.
- ❖ There are some other thermodynamic variables such as entropy, internal energy, etc., but these thermodynamic variables can be expressed in terms of pressure, volume and temperature.

- The present status of the system described in terms of properties such as pressure, temperature, and volume is called the state of system.

Fundamental Gas Laws

Ideal Gas Law ($PV = nR^*T$)

- Fundamental in understanding the behavior of the atmosphere.
- Explains how pressure, temperature, and volume are interrelated.
- Used in deriving the **hydrostatic equation**, which describes how atmospheric pressure decreases with height.
- Helps in weather prediction models by explaining air density and how it affects buoyancy, leading to cloud formation and storm development.



$$P = \rho RT$$

Where:

P = pressure in Pa

ρ = Density = mass(kg)/Volume(m^{-3}) = (m/V)

R = the Gas constant; with its value depend on the type of gas under consideration.

T = the temperature in degree Kelvin ($\text{K} = 273.15 + ^\circ\text{C}$).

Cont. . .

- ❖ How much gas is present could be specified by giving the mass instead of the chemical amount of gas. Therefore, an alternative form of the ideal gas law may be useful.

$$n = m/M$$

- ❖ Where, n is the amount of the gas (in moles), m is the total mass of the gas (in grams) and M is the molar mass of the gas (in grams per mole)

Cont. . .

- ❖ By replacing n with m/M and subsequently introducing density $\rho = m/V$, we get:

$$PV = (m/M)R^* T$$

R^ is called the universal gas constant*

$$P = \rho(R^*/M)T$$

The magnitude of R^ is 8.3145 J/ K mol*

- ❖ Defining the specific gas constant R as the ratio R^*/M ,

$$P = \rho RT$$

- ❖ Alternatively, the law may be written in terms of the specific volume (α), the reciprocal of density ($1/\rho$), as

$$P\alpha = RT$$

Gas law for dry air and water vapor

❖ Using subscripts d and v for dry air and water vapor, respectively, as follows

$$P_d = \rho_d R_d T$$

$$e = P_v = \rho_v R_v T$$

✓ Where R_d and R_v are the gas constant for 1 kg of dry and moist air respectively.

❖ We can define the apparent molecular weight M_d of dry air as the total mass (in grams).

$$M_d = \frac{\sum_i m_i}{\sum_i \frac{m_i}{M_i}} = 28.97 \text{ g/mol}$$

(M_v (molecular weight of H_2O) = 18.016).

Cont. . .

- ❖ Therefore the dry gas constant for 1kg of dry air calculated as follows;

$$R_d = 1000 \frac{R}{M_d} = 1000 \frac{8.3145}{28.97} = 287 JK^{-1} Kg^{-1}$$

$$R_v = 1000 \frac{R}{M_w} = 1000 \frac{8.3145}{18.016} = 461.51 JK^{-1} Kg^{-1}$$

- ❖ So that

$$M_v < M_d \implies R_v > R_d$$

- ❖ The ratio of gas constants for dry air and water vapor is calculated as :

$$\varepsilon \equiv \frac{R_d}{R_v} = \frac{M_v}{M_d} \approx 0.622$$

Example 1

A weather balloon measures an air temperature of 250 K and an atmospheric pressure of 500 hPa (or 50,000 Pa) at a certain altitude. Assuming the air behaves as an ideal gas, calculate the air density at that altitude.

Example 2

- At what temperature in degree Celsius will water vapor of density $6.67 \times 10^{-3} \text{ kg/m}^3$ exerted a pressure of 900 pa ?

Example 3

What is the mass of dry air occupying a room of dimensions $3 \times 5 \times 4$ m at $p = 1$ atmosphere and $T = 20^\circ\text{C}$?

....END

Thanks !

Atmospheric Thermodynamics

Chapter 1: Introduction



Objective

- To explain Fundamental Gas Laws
 - Avogadro's law
 - Boyle's Law
 - Charles's Law
 - Dalton's law
- To derive Virtual Temperature

Avogadro's Hypothesis

- "Equal volumes of gases at the same temperature and pressure contain an equal number of molecules."
- This implies that the volume of a gas is directly proportional to the number of moles at constant temperature and pressure.

- Important for understanding the mixing and diffusion of atmospheric gases, such as oxygen, nitrogen, and greenhouse gases.
- Used in calculating the molecular weight of air, which is essential for atmospheric studies.

Cont. . .

❖ Avogadro's law describes the relationship between the volume and the amount of a gas:

- ✓ At constant temperature and pressure, the volume of a sample of gas is directly proportional to the number of moles of gas in the sample. Stated mathematically,

$$V \propto n$$

at constant T and P

❖ We can demonstrate the relationship between the volume and the amount of a gas by filling a balloon; as we add more gas, the balloon gets larger.

Cont. . .

- Alternative Form (Comparing Two States): If the amount of gas changes from n_1 to n_2 while temperature and pressure remain constant, then:

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

- If you increase the number of gas molecules (n), the volume increases.
- If you decrease the number of gas molecules, the volume decreases.
- This law explains why gases expand when more particles are added under the same conditions.

Example

- A balloon is filled with 0.5 moles of helium gas, occupying a volume of 10 liters at constant temperature and pressure. If 1 more mole of helium is added, what will be the new volume of the balloon?

Boyle's Law ($P_1V_1 = P_2V_2$ at constant T)

- Helps explain how air compresses and expands in the atmosphere.
- Relevant in **adiabatic processes**, such as when rising air expands and cools, leading to cloud formation and precipitation.
- Important for understanding the behavior of air masses and pressure systems in weather patterns.

Cont. . .

$$P_1V_1=P_2V_2$$

- means that the product of pressure and volume remains constant.
- If pressure increases, then volume must decrease to keep the product the same.
- If volume increases, then pressure must decrease.

Example

- A weather balloon is released from sea level where the atmospheric pressure is 1000 hPa and the volume of the balloon is 2 m³. As it rises to an altitude where the pressure drops to 500 hPa (0.5 atm), what will be the new volume of the balloon, assuming temperature remains constant?

Cont. . .

- Before a thunderstorm, a pocket of air at an altitude has a volume of 3 m^3 and a pressure of 800 hPa. As the storm moves in, the descending air compresses the air pocket to 2 m^3 . What is the new pressure?

Charles's Law

❖ At constant pressure, the volume of a fixed amount of gas is directly proportional to its absolute temperature (in kelvins).

❖ Charles's law and is stated mathematically as

$$V \propto T$$

❖ Charles's law is valid for virtually all gases at temperatures well above their boiling points.

❖ Hot air rises, which is why hot-air balloons ascend through the atmosphere and why warm air collects near the ceiling and cooler air collects at ground level

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

where:

- V_1, V_2 = Initial and final volumes of the gas
- T_1, T_2 = Initial and final temperatures (in Kelvin)

Dalton's Law of Partial Pressures ($P_{\text{total}} = P_1 + P_2 + \dots$)

- Dalton's Law deals with a mixture of ideal gases. It states that the *partial pressures* of the individual constituent gaseous components are additive.
- Crucial for understanding the composition of the atmosphere and how gases interact.

Cont. . .

- Helps in calculating the **vapor pressure of water**, which is essential for understanding humidity and dew point.
- Important in explaining the formation of clouds, fog, and precipitation based on water vapor content

Cont. . .

- ❖ We thus conclude that the total pressure is the sum of the partial pressures.
- ❖ This expresses Dalton's law which states that for a mixture of K components,

$$P = \sum_{i=1}^k P_i$$

Exercise

- If at 0 °C the density of dry air alone is 1.275 kg/m³ and the density of water vapor alone is 4.770×10^{-3} kg/m³, what is the total pressure exerted by a mixture of the dry air and water vapor at 0 °C?

Mixture of dry air and vapor

- ❖ From Dalton's Law of Partial Pressures, the total pressure p for a mixture of dry air and vapor is:

$$P = P_d + e$$

- ❖ Thus, the ideal gas law for the mixture of dry air and vapor is:

$$P = (\rho_d R_d + \rho_v R_v) T$$

- ❖ We can express the density of the mixture as:

$$\rho = \rho_d + \rho_v \equiv (m_d + m_v) / V$$

Applications of Gas laws in Meteorology

Weather forecasting:

- ✓ Numerical weather models use the ideal gas law to simulate atmospheric conditions.

Cloud formation:

- ✓ Boyle's law and Dalton's law help explain condensation and precipitation processes.

Cont. . .

- **Storm development:**

- ✓ The expansion and cooling of rising air masses contribute to thunderstorm formation.

- **Atmospheric circulation:**

- ✓ Gas laws explain pressure differences that drive winds and global weather patterns.

Virtual Temperature

Virtual Temperature

- Is Apparent temperature of moist air, accounting for the presence of water vapor.
- The temperature that dry air would need to have in order to match the density of moist air at the same pressure. This is done because moist air is less dense than dry air at the same temperature.
- Used to adjust buoyancy and stability in meteorological calculations involving moist air.

The Difference between actual and virtual Temperature

Actual Temperature	Virtual Temperature:
Measured temperature of the air	This is the apparent temperature of moist air that accounts for the presence of water vapor
Directly measured	$T_v = \frac{T}{1 - e/p(1 - \epsilon)}$
Kinetic energy of air molecules	Corrected temperature that accounts for buoyancy of moist air

Why is the virtual temperature always greater than the actual temperature?

- Water Vapor is Lighter: Water vapor molecules are less dense than dry air molecules, making moist air less dense. Virtual temperature accounts for this by being higher.
- Increased Buoyancy: Water vapor makes moist air more buoyant. To balance this, virtual temperature is adjusted to a higher value.
- Moist Air Appears Warmer: Moist air, being less dense, behaves as if it's warmer than dry air at the same actual temperature, so virtual temperature is higher.

Cont. . .

- The virtual temperature equation is important in meteorology because it helps model the buoyancy of moist air, which directly affects weather phenomena like convection, cloud formation, and precipitation.
- By deriving and using this equation, meteorologists can more accurately predict weather patterns and better understand the impact of water vapor on atmospheric dynamics.
- The virtual temperature is a corrected value of the actual temperature, accounting for the presence of water vapor, which influences air density and buoyancy

Cont. . .

- ❖ The density of the mixture of dry air and water vapour is

$$\rho = \rho_d + \rho_v$$

- ❖ Dalton's law of partial pressure, $p = p_d + e$, where

$$P_d = \rho_d R_d T$$

$$e = P_v = \rho_v R_v T$$

- ❖ Combining the above equations, and considering $\mathbf{p = RdpTv}$ we can calculate virtual temperature

$$T_V = \frac{T}{1 - (e/p)(1 - \varepsilon)}$$

Refer to the detailed step-by-step derivation on the classroom board.

- ❖ Where T: actual temperature

p: actual (total) pressure = $p_d + e$

p_d : partial pressure exerted by dry air

e: partial pressure exerted by water vapor

$$\varepsilon = R_d/R_v = 0.622$$

- ❖ Then the ideal gas law can be written in terms of virtual temperature as : $\mathbf{p = RdT_v}$

Example

- You are working on weather forecasting, and you need to determine the virtual temperature of the air in a region with the following conditions. Actual temperature of 25°C , Vapor pressure 15 hPa., total pressure 1010 hPa.

Atmospheric Thermodynamics

Chapter 1: Introduction



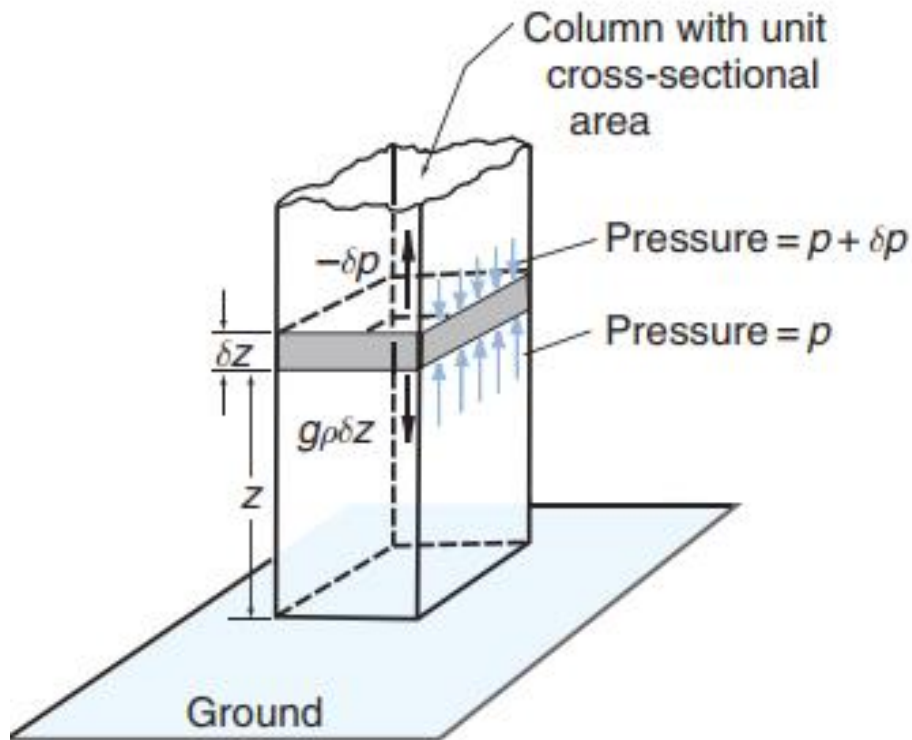
Objective

- To Derive Hydrostatic equation
- To Know the importance of Hydrostatic equation
- To Explain Geopotential height
- To Derive Hypsometric equations

The Hydrostatic Equation

- ❖ The hydrostatic equation is one of the most fundamental equations in atmospheric thermodynamics and meteorology.
- ❖ It explains how pressure changes with height in the atmosphere and forms the basis of many meteorological calculations such as
 - ❑ altitude determination
 - ❑ atmospheric structure
 - ❑ Determines Geopotential Height

- ❖ If the net upward pressure force on the slab is equal to the downward force of gravity on the slab, the atmosphere is said to be in **hydrostatic balance**.



- downward gravitational force = $g\rho\delta z$
- Since we know that pressure decreases with height, δp must be a *negative* quantity ($-\delta p$)

Fig: Balance of vertical forces in an atmosphere in which there are no vertical accelerations (i.e., an atmosphere in hydrostatic balance)

- ❖ For an atmosphere in hydrostatic equilibrium, there is a balance of forces in the vertical, i.e.,

$$-\delta p = g\rho\delta z$$

- ❖ In the limit as $\delta z \rightarrow 0$,

$$\partial p / \partial z = -g\rho$$

- ✓ *This is the hydrostatic equation.*

- ✓ Refer to the detailed step-by-step derivation on the classroom board.

- ❖ The negative sign ensures that the *pressure decreases with increasing height.*

Cont. . .

❖ Since $\rho = 1/\alpha$, the equation can be rearranged to give

$$g \, dz = -\alpha \, dp$$

❖ Integrating the hydrostatic equation from height z (and pressure $p(z)$) to an infinite height:

$$-\int_{p(z)}^{p(\infty)} dp = \int_z^{\infty} g \rho dz$$

❖ Since $p(\infty) = 0$,

$$\mathbf{P(z) = \int_z^{\infty} g \rho dz}$$

❖ That is, the pressure at height z is equal to the weight of the air in the vertical column of unit cross-sectional area lying above that level.

Importance in Meteorology

- Explains why atmospheric pressure **decreases** with height.
- Governs large-scale atmospheric circulation and weather patterns.
- Helps in predicting the structure of the atmosphere and pressure variations.

Exercise

Given the following information

- Surface pressure (P_0) = 1013 hPa (or 101,300 Pa)
- Density of air (ρ) = 1.225 kg/m³ (at sea level)
- Gravity (g) = 9.81 m/s²

Find change in pressure for $z = 1\text{km}$ and 2km

Geopotential (Φ)

- Geopotential is defined as the work(energy) required to lift a unit mass of air from sea level to a given height against Earth's gravity.

Cont. . .

- ❖ The geopotential Φ at any point in the Earth's atmosphere is defined as the work that must be done against the Earth's gravitational field to raise a mass of 1kg from sea level to that point.
- ❖ The units of geopotential are J/kg or m^2/s^2 . The work (in joules) in raising 1 kg from z to $z + dz$ is gdz . Therefore

$$d\Phi = gdz$$

or, using the hydrostatic equation,

$$d\Phi = gdz = -\alpha dp$$

Cont. . .

❖ The geopotential $\Phi(z)$ at height z is thus given by

$$\Phi(z) = \int_0^z g \, dz$$

✓ where the geopotential $\Phi(0)$ at sea level ($z = 0$) has been taken as zero.

❖ The geopotential at a particular point in the atmosphere depends only on the height of that point and not on the path through which the unit mass is taken in reaching that point.

Geopotential height (Z)

❖ We define the geopotential height Z as

$$Z = \frac{\Phi(z)}{g_0} = \frac{1}{g_0} \int_0^z g dz$$

✓ where g_0 is the globally averaged acceleration due to gravity at the Earth's surface (9.81 m/s²).

❖ Geopotential height is used as the vertical coordinate in most atmospheric applications in which energy plays an important role.

❖ Geopotential height can be seen from the Table below that the values of Z and z are almost the same in the lower atmosphere where $g \approx g_0$.

Values of the geopotential height (Z), and acceleration due to gravity (g), at 40° latitude for geometric height (z)

z (Km)	Z (Km)	g (m/s ²)
0	0	9.81
1	1	9.8
10	9.99	9.77
100	98.47	9.5
500	463.6	8.43

Difference Between Geopotential Height and Geometric Height

Aspect	Geopotential Height (Z)	Geometric Height (z)
Definition	A height measure that accounts for the variation in Earth's gravity.	The actual physical height above a reference surface (usually sea level)
Formula	$Z = \frac{1}{g_0} \int_0^z g dz$	z is simply the measured altitude
Gravity Consideration	Takes into account that gravity (g) decreases with altitude.	Assumes a fixed reference, not considering gravity variations.
Typical Units	Geopotential meters (gpm)	Meters (m)
Key Concept	A corrected height scale that accounts for gravity variation.	Measures the direct vertical distance above the Earth's surface

Why is considering gravity important?

- If we use geometric height without correction, it overestimates the actual energy needed to lift air.
- Since work depends on both height and gravity, we must account for gravity changes.
- Many equations (e.g., hydrostatic balance, thermal wind) work better when using geopotential height, since it reflects the true energy state of air parcels.

Hypsometric Equation

- The Hypsometric equation relates the thickness of an atmospheric layer to its mean temperature and pressure difference.
- It describes how pressure changes with height in a stratified atmosphere.

$$Z_2 - Z_1 = \frac{RT_v}{g} \ln \left(\frac{P_1}{P_2} \right)$$

where:

- $Z_2 - Z_1$ = thickness of the atmospheric layer
- R = specific gas constant for dry air
- T_v = virtual temperature (accounts for water vapor)
- g = gravitational acceleration
- P_1, P_2 = pressure at different heights

- ❖ In meteorological practice it is not convenient to deal with the density of a gas, ρ , the value of which is generally not measured.
- ❖ By making use of $(p = \rho RT)$ to eliminate ρ in $(\frac{\partial p}{\partial z} = -g\rho)$, we obtain

$$\frac{\partial p}{\partial z} = \frac{-pg}{RT} = \frac{-pg}{R_d T_v}$$

❖ Rearranging the last expression and using $d\Phi = g \, dz$ yields

$$d\Phi = g \, dz = -\alpha \, dp$$

$$d\Phi = g \, dz = -RT \frac{dp}{p} = -R_d T_v \frac{dp}{p}$$

❖ If we now integrate between pressure levels p_1 and p_2 , with Geopotential Φ_1 and Φ_2 , respectively,

$$\int_{\Phi_1}^{\Phi_2} d\Phi = - \int_{p_1}^{p_2} R_d T_v \frac{dp}{p}$$

$$\Phi_2 - \Phi_1 = -R_d \int_{p_1}^{p_2} T_v \frac{dp}{p}$$

❖ Dividing both sides of the last equation by g_0 and reversing the limits of integration yields

$$Z_2 - Z_1 = \frac{Rd}{g_0} \int_{p_2}^{p_1} T_v \frac{dp}{p}$$

❖ This difference $Z_2 - Z_1$ is called the Geopotential ***thickness of the layer*** between pressure levels p_1 and p_2 .

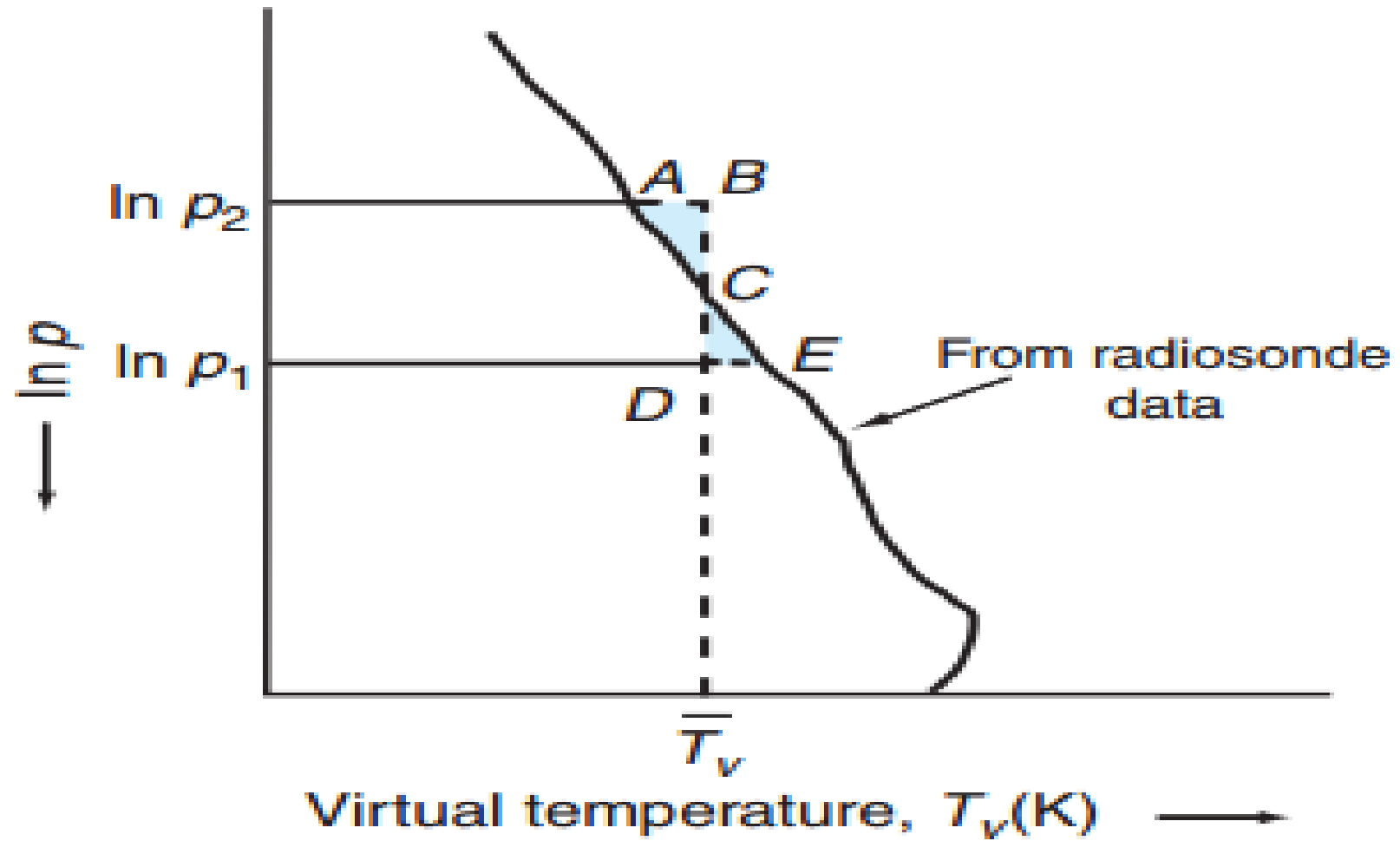
- ❖ For an isothermal atmosphere (i.e., temperature constant with height), if the virtual temperature correction is neglected, (above equation) becomes:

$$Z_2 - Z_1 = H \ln \left(\frac{p_1}{p_2} \right)$$

$$p_2 = p_1 e^{\left[\frac{-(Z_2 - Z_1)}{H} \right]}$$

- ❖ Where $H = \frac{R_d T_v}{g_0}$ is the *scale height*. Since $R_d = 287 J K^{-1} kg^{-1}$ and $g_0 = 9.81 m s^{-2}$ we have, approximately, $H = 29.3 T_v$.
- ❖ If we take a *mean* value for virtual temperature of $T_v = 255$ K, the scale height H for air in the atmosphere is found to be about 7.5 km.

Cont. . .



- ❖ The temperature and vapour pressure of the atmosphere generally vary with height and virtual temperature correction cannot always be neglected.
- ❖ In this case we can define a mean virtual temperature $\bar{T}_v$ (see the above Figure)

$$Z_2 - Z_1 = \bar{H} \ln \left(\frac{p_1}{p_2} \right) = \frac{R_d \bar{T}_v}{g_o} \ln \left(\frac{p_1}{p_2} \right)$$

- ❖ ***This is the Hypsometric equation***

Refer to the detailed step-by-step derivation on the classroom board.

Why is Atmospheric Layer Thickness Important?

- The **thickness** of an atmospheric layer refers to the vertical distance between two pressure levels in the atmosphere. This is important for

Weather Forecasting:

- Thickness is related to temperature: warmer air expands, making layers thicker, while colder air contracts, making them thinner.

Aviation:

- Helps in determining altitude-pressure relationships, crucial for aircraft navigation

Example

Calculate the thickness of the layer between the 1000 and 500hPa pressure surfaces (a) at a point in the tropics where the mean virtual temperature of the layer is 15°C and (b) at a point in the polar regions where the corresponding mean virtual temperature is -40°C .

....END

Thanks !